Nautilus - Presentation & Demo Guide

How to Present the Optimal Ship Routing Project

A step-by-step guide: pitch, slides, live demo and viva answers

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1 How to Use This Guide

This guide walks you through presenting the project from start to finish: a short pitch, a suggested slide deck with talking points, a click-by-click live-demo script, a numbers cheat-sheet, and strong answers to the questions examiners usually ask. Read it once, do one practice run with the demo, and you are ready.

2 30-Second Elevator Pitch

‘Ships still plan routes on static charts that ignore today's weather, which wastes fuel and risks safety. Our system, Nautilus, models the sea around India as a grid using real depth data, keeps ships in deep water and 22 km off the coast, and finds the best route with a weather-aware A* search. A small machine-learning model predicts how much waves slow the chosen ship, so the route adapts to the vessel and the sea state — even live weather. It runs free and offline, and we show it on an interactive Google-Maps app.’

3 Suggested Slide Deck (11 slides)

#	Slide	What to say / show
1	Title	Project title, your names, guide, project code 43.
2	Problem	Static routing ignores weather → more fuel, cost, emissions, risk.
3	Objective	A fast, safe, adaptive, *free* routing system for Indian ports.
4	Data	GEBCO depth + ERA5 waves + live Open-Meteo (show the dataset table).
5	Method overview	The 7-stage pipeline diagram (preprocess → grid → ML → routing → UI).
6	Grid & safety	Depth mask, 22 km coast buffer, KD-tree snapping.
7	Algorithms	Dijkstra baseline vs weather-aware A*; admissible heuristic.
8	ML layer	Random Forest predicts ship speed from waves; feeds A* cost.
9	Live demo	Switch to the app (see the demo script in section 4).
10	Results	74% fewer cells; +298 km to cut peak wave 4.13→3.31 m; ≥22 km coast.
11	Conclusion + future	Free, offline, laptop-safe; future: currents, AIS training, forecast slider.

4 Live Demo Script (click-by-click)

Have the app running **before** you start talking. In a terminal, in the project folder:

```
python server.py
```

Then open **http://localhost:8000**. Now walk through:

- **Step 1 — Set the scene.** 'This is Nautilus. On the left are the controls; the map shows the seas around India with a live wave overlay.'
- **Step 2 — Pick a vessel.** Choose *General Cargo* (it reacts most to weather, so the demo is clearer). Point out the service-speed / draft line.
- **Step 3 — Pick the voyage.** Origin *Cochin (Kerala)*, destination *Dubai / Jebel Ali (UAE)*.
- **Step 4 — Choose weather.** Click **Rough** to show a stormy scenario (the effect is dramatic).
- **Step 5 — Find route.** Click **Find optimal route**. Point to the two lines: blue = Dijkstra (shortest), red = weather-aware A*.
- **Step 6 — Tell the story.** Read the verdict card aloud: A* takes a slightly longer path to avoid the roughest water, and explores far fewer cells.
- **Step 7 — Show detail.** Click a coloured dot on the route to show the wave tooltip (height / period / direction). Point out the ship icon at the origin and the anchor at the destination.
- **Step 8 — Go live.** Click ● **Live**, run the route again — 'now it uses real-time waves from a free API.'
- **Step 9 — Change the ship.** Switch to *Container Ship* and re-run — the bigger ship powers through, so its route barely bends. This proves the ML is vessel-specific.

Tip: if the room has no internet, skip Step 8 and stay on Typical/Rough — those are fully offline.

5 Numbers to Remember (cheat-sheet)

Fact	Value
Region	lat 0–27°N, lon 50–100°E
Grid	540×1000 cells @ 0.05° (~5.5 km); 337,196 navigable
Coast safety buffer	22 km
Raw data shrunk	11.6 GB → ~1 MB working data
ML model	Random Forest; R^2 = 0.998; error 0.17 knots; ~11 MB
A* efficiency	~74% fewer cells explored than Dijkstra
Flagship result	Cochin→Dubai: +298 km to cut peak wave 4.13→3.31 m
Cost	Zero (all free / open-source)

6 Likely Questions & Strong Answers

Q: Where is the machine learning in this project?

A: A Random Forest model predicts the ship's attainable speed from the sea state and vessel parameters. The A* search uses that prediction as its cost (time = distance ÷ predicted speed), so the ML directly drives the routing decision. It is the 'ML-based decision technique' from our synopsis.

Q: Why not deep learning or reinforcement learning?

A: They need GPUs, large datasets and hours of training, and their decisions are hard to explain. A Random Forest trains in seconds on a CPU, is very accurate here ($R^2 = 0.998$) and is fully explainable — a better fit for a free, laptop-based, defensible system.

Q: You trained on synthetic data — is that valid?

A: Yes. Real per-ship speed logs are not public, so we generated training data from an established added-resistance relation (wave resistance grows with wave height squared and is worst in head seas) plus noise. The model learns that physical relationship, and it can be retrained on real AIS data later with no code change.

Q: Why A* instead of just Dijkstra?

A: A* uses a great-circle heuristic to head toward the goal, so it explores about 74% fewer cells while still returning the optimal route (the heuristic is admissible, which guarantees optimality). We keep Dijkstra as the traditional baseline to prove the improvement.

Q: Why GEBCO and ERA5 specifically?

A: GEBCO is the best free global depth dataset — fine enough to resolve coastlines and shoals, where nautical charts are paid and ETOPO is too coarse. ERA5 gives wave height, direction and period together, for free — exactly the three inputs that decide how hard a passage is.

Q: How does it stay free and not hang the laptop?

A: Every tool and dataset is free and open-source; live weather uses Open-Meteo which needs no key or card. The giant files are streamed lazily and shrunk once to ~1 MB, and per-move costs are pre-computed, so a route solves in 1–2 seconds without heavy memory use.

Q: What are the safety guarantees?

A: A cell is navigable only if the water is deep enough and at least 22 km from the coast. Ports snap to the nearest safe cell in the main ocean body, so a route is always reachable and always keeps clear of land and shallow water.

Q: Is this real-time?

A: It can be. The 'Live' scenario pulls current waves from the Open-Meteo Marine API and routes on them. For offline demos we also ship two stored scenarios, Typical and Rough.

Q: What is the limitation / future scope?

A: It currently covers the seas around India and uses waves (not currents). Future work: widen the region, add ocean currents and traffic-density avoidance, a forecast time-slider, and retraining the ML model on real AIS voyage data.

7 Presentation Tips

- **Do** run the app once before the session and keep the terminal open.

- **Do** lead with the problem and the live demo — they are the most convincing parts.
- **Do** use the Rough scenario with General Cargo for the clearest visual difference.
- **Don't** rely on internet for the demo unless you have tested it in the room; keep Typical/Rough as a safe fallback.
- **Don't** over-claim the ML — describe it honestly as a speed predictor that guides the search; that is exactly what examiners respect.
- **Close** with the one-liner: ‘A free, offline, explainable system that makes ship routes safer, calmer and more efficient.’